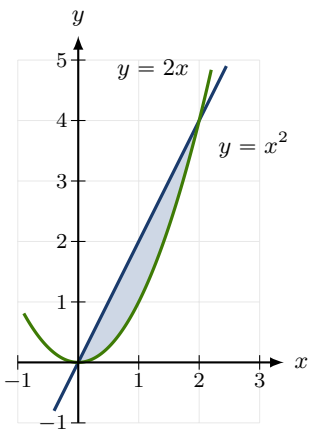
L4.2b Washers

Work in your group. Check your answers against the Quarto tonight.

Washers

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1. The region $\mathcal{R}$ enclosed by $y = 2x$ and $y = x^2$ is rotated about the x-axis.



(a) Draw a representative washer dV and label its radii $R(x)$ and $r(x)$. Find where the curves meet: the path and limits.

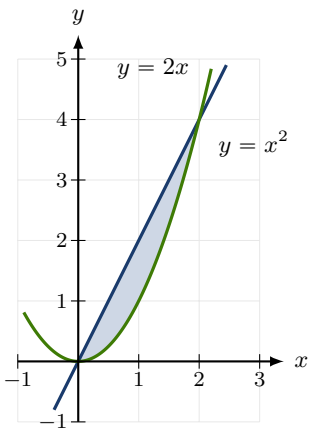
(b) Test a point to decide which curve is the outer curve and which is the inner curve. Write $R(x)$ and $r(x)$.

(c) Write $dV = A(x)\,dx$.

(d) Find the volume of the solid.

Rotating about other lines

2. The region $\mathcal{R}$ enclosed by $y = 2x$ and $y = x^2$ is rotated about the line $y = 4$.

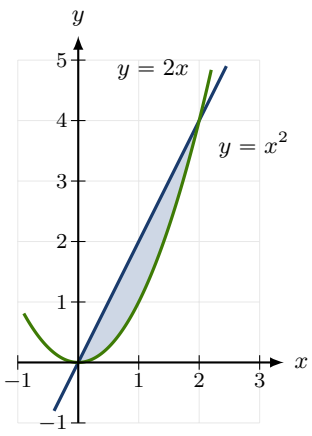


(a) Draw the axis of rotation, a representative washer dV, and its radii $R(x)$ and $r(x)$, measured from the axis.

(b) Write $R(x)$, $r(x)$ and $dV = A(x)\,dx$.

(c) Find the volume of the solid.

3. The region $\mathcal{R}$ enclosed by $y = 2x$ and $y = x^2$ is rotated about the line $y = -1$.

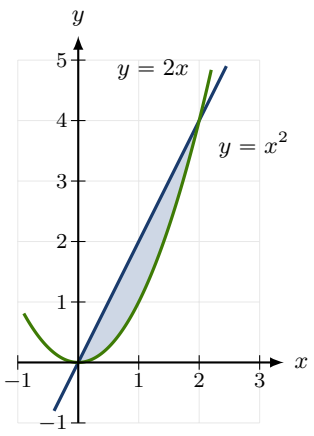


(a) Draw the axis of rotation, a representative washer dV, and its radii $R(x)$ and $r(x)$, measured from the axis.

(b) Write $R(x)$, $r(x)$ and $dV = A(x)\,dx$.

(c) Find the volume of the solid.

4. The region $\mathcal{R}$ enclosed by $y = 2x$ and $y = x^2$ is rotated about the line $x = 2$.



(a) Draw the axis of rotation, a representative washer dV, and its radii $R(y)$ and $r(y)$. Test a value of y to decide which curve is on the right.

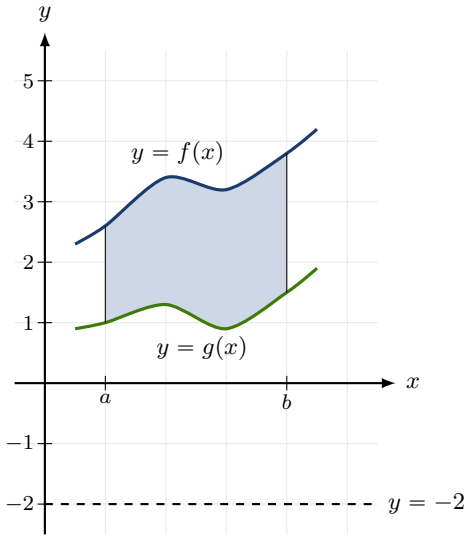
(b) Path and limits: solve each curve for x, and find the limits in y.

(c) Write $R(y)$, $r(y)$ and $dV = A(y)\,dy$.

(d) Find the volume of the solid.

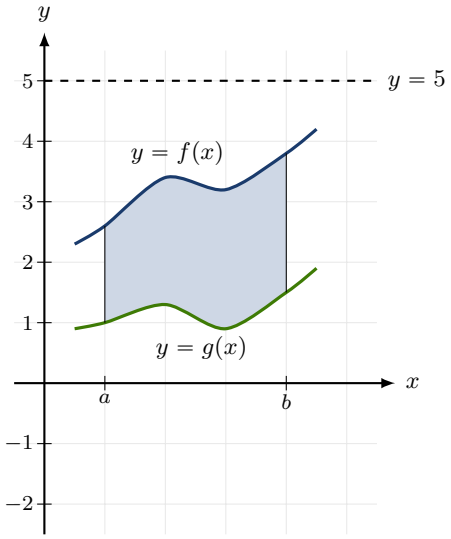
5. The region between $y = f(x)$ and $y = g(x)$ on $[a, b]$ is rotated about the dashed line. On each graph, draw a representative washer, label $R(x)$ and $r(x)$, and write $A(x)$ in terms of f and g.

(a)



(a) $A(x) =$

(b)



(b) $A(x) =$